

**CHANDIGARH UNIVERSITY
APEX INSTITUTE OF TECHNOLOGY**



ASSIGNMENT 1

Subject: Computer Networks

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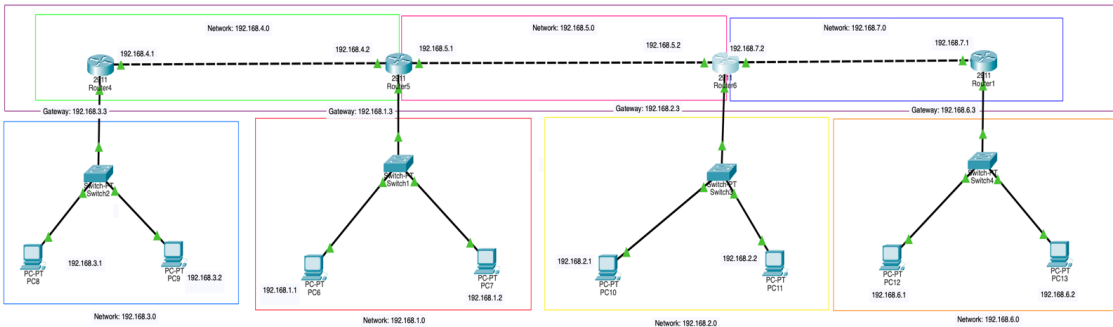
Mr. Alok Singh Rathore

Question 1

Set up a static-routing network consisting of 4 routers, 4 switches and 8 PCs. Clearly mark out each network along with its Network ID, and list the IP addresses in plain text alongside the diagram.

Answer:

The topology below was built with four routers (Router1, Router4, Router5 and Router6) linked back-to-back over serial WAN connections, and four switches, each feeding two PCs on its own LAN segment — giving eight end devices in total. Static routes were then configured on every router so that traffic could reach the three networks it isn't directly attached to.



Static Routing Table:

Router	Destination Network	Subnet Mask	Next Hop
Router4	192.168.1.0	255.255.255.0	192.168.4.2
Router4	192.168.2.0	255.255.255.0	192.168.4.2
Router4	192.168.6.0	255.255.255.0	192.168.4.2
Router5	192.168.3.0	255.255.255.0	192.168.4.1
Router5	192.168.2.0	255.255.255.0	192.168.5.2
Router5	192.168.6.0	255.255.255.0	192.168.5.2
Router6	192.168.1.0	255.255.255.0	192.168.5.1
Router6	192.168.3.0	255.255.255.0	192.168.5.1
Router6	192.168.6.0	255.255.255.0	192.168.7.1
Router1	192.168.2.0	255.255.255.0	192.168.7.2
Router1	192.168.1.0	255.255.255.0	192.168.7.2
Router1	192.168.3.0	255.255.255.0	192.168.7.2

Network Table:

Network ID	Purpose	Devices Connected
192.168.1.0	LAN	Router5, PC6, PC7
192.168.2.0	LAN	Router6, PC10, PC11
192.168.3.0	LAN	Router4, PC8, PC9
192.168.4.0	WAN	Router4 to Router5
192.168.5.0	WAN	Router5 to Router6
192.168.6.0	LAN	Router1, PC12, PC13
192.168.7.0	WAN	Router6 to Router1

Question 2

Tabulate the differences between a Hub, a Switch and a Router. Also justify why, among the three, a Router is the smartest device and a Hub is the dumbest.

Answer:

Hubs, switches and routers all move data around a network, but they don't do it the same way — each sits at a different layer of the OSI model and makes a different amount of 'thinking' before it forwards a frame or packet. The comparison below lays out how they differ.

Hub vs Switch vs Router

Parameter	Hub	Switch	Router
Layer of Operation	Physical Layer (L1)	Data Link Layer (L2)	Network Layer (L3)
Addressing Used	None	MAC Address	IP Address
How Data Moves	Broadcasts to every port	Sent only to the target device	Routed across separate networks
Intelligent Forwarding?	No	Yes — via a MAC address table	Yes — via a routing table
Collision Domain	Single, shared	One per port	One per interface
Broadcast Domain	Single	Single (no VLANs)	Separate per interface
Relative Speed	Slowest	Fast	Fastest (route-dependent)
Security Level	Minimal	Moderate	Strongest
Typical Role	Wiring devices together in a LAN	Efficient switching inside a LAN	Linking distinct networks together

Why does the Router come out on top?

- It operates at Layer 3, so it reasons in terms of IP addresses rather than raw wiring.
- It builds and consults a routing table to work out the best path for each packet.
- It's the device that actually joins separate networks — LANs to WANs, and LANs to each other.
- Because it forwards packets only toward the correct destination network, it keeps unnecessary traffic off segments that don't need it.

Why is the Hub at the bottom of the list?

- It only understands electrical signals — Layer 1 — with no concept of addressing.
- It has no MAC or IP awareness whatsoever.
- Every incoming frame is repeated out of all other ports, destination or not.
- With no forwarding logic, it can't filter traffic, so collisions and congestion pile up quickly.

In short: a Hub is a dumb signal repeater, a Switch adds intelligence by learning MAC addresses to deliver frames only where they're needed, and a Router goes a step further by using IP addressing and routing tables to connect entire networks and choose the best path for traffic — which is exactly why it sits at the top of the intelligence ladder and the Hub at the bottom.

Question 3

What is meant by Random Access Protocols, and why are they needed? Describe each category with a diagram and an example, and also explain how CSMA/CD differs from CSMA/CA.

Answer:

On a shared network link, several devices may try to use the same channel at once. When two transmissions overlap in time, the signals collide and the data is lost. Random Access Protocols exist to manage exactly this situation — they are Medium Access Control (MAC) schemes that let a station transmit whenever it has data ready, while laying down rules for sensing the channel and recovering from collisions when they happen.

Why are Random Access Protocols needed?

- They let many devices share one physical channel.
- They cut down on the number of collisions.
- They keep the network running efficiently.
- They give every device a fair shot at the medium.
- They avoid the wasted retransmissions that would otherwise clog the network.

Categories of Random Access Protocols

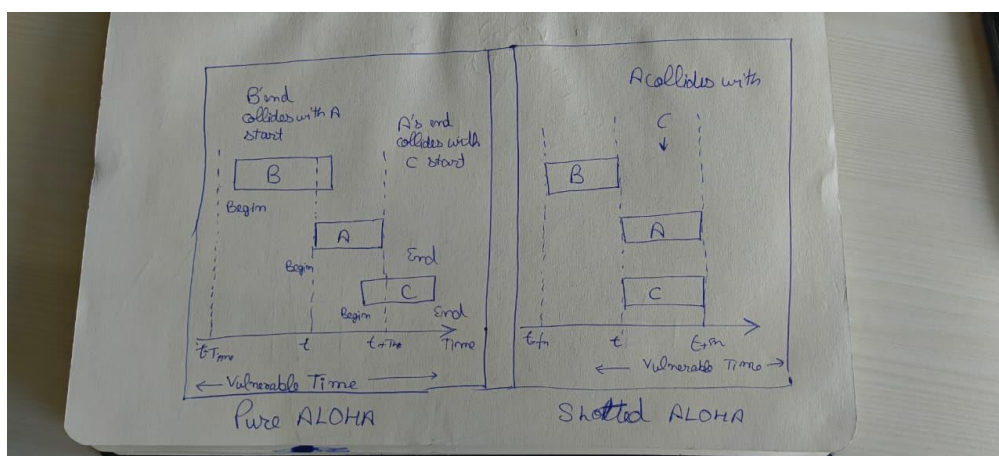
- ALOHA
- CSMA (Carrier Sense Multiple Access)
- CSMA/CD (Collision Detection)
- CSMA/CA (Collision Avoidance)

1. ALOHA

ALOHA is the earliest and most basic random access scheme. A station simply sends its frame the moment it has data — it never bothers checking whether the channel is already in use. If two stations happen to transmit in overlapping time windows, a collision wipes out both frames, and each sender must back off for a random interval before trying again.

Working:

- A station transmits as soon as it has data, with no prior check.
- Receiving an ACK means the frame got through cleanly.
- No ACK arriving is treated as a sign that a collision happened.
- The sender then waits a random backoff period and retransmits.



Example:

If Computer A and Computer B both happen to transmit at the same moment, their signals overlap and collide. Neither gets through, so both wait a random amount of time before resending.

Variants of ALOHA:

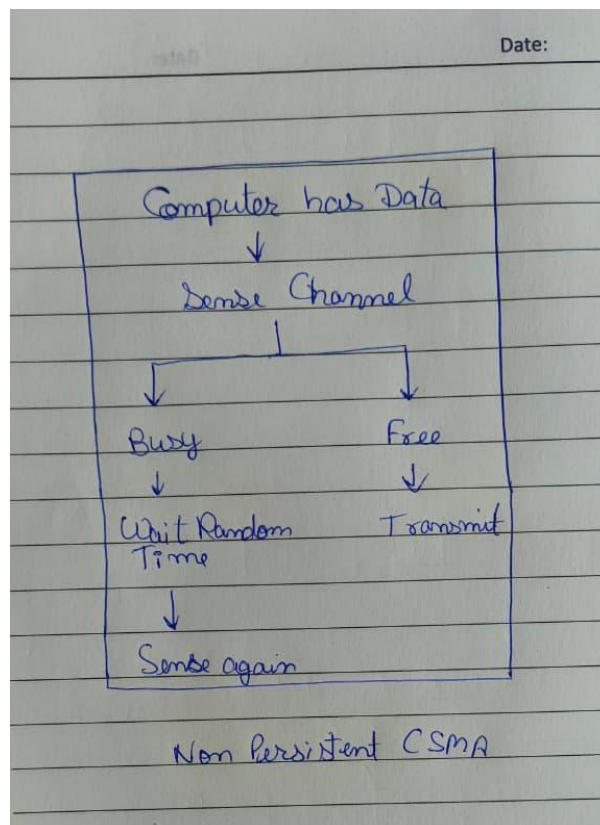
- Pure ALOHA
- Slotted ALOHA

2. CSMA (Carrier Sense Multiple Access)

CSMA improves on plain ALOHA by having the station listen to the channel before it sends anything. A free channel means the station goes ahead and transmits; a busy one means it holds off and checks again later.

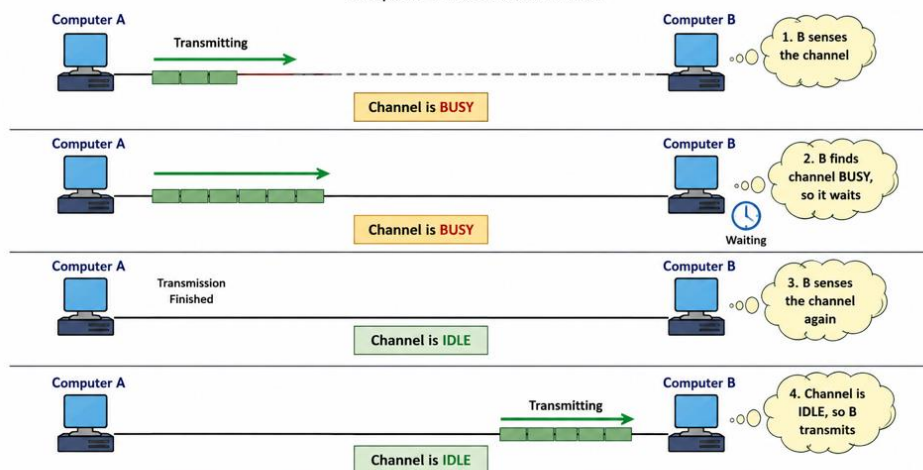
Working:

- Sense the channel first.
- Transmit immediately if it's idle.
- Hold off if it's busy.
- Re-check the channel once it frees up and try again.



CSMA (Carrier Sense Multiple Access)

Scenario: Computer A is already transmitting.
Computer B wants to send data.



In CSMA, a station listens to the channel before transmitting. If the channel is busy, it waits. If the channel is idle, it transmits.

Example:

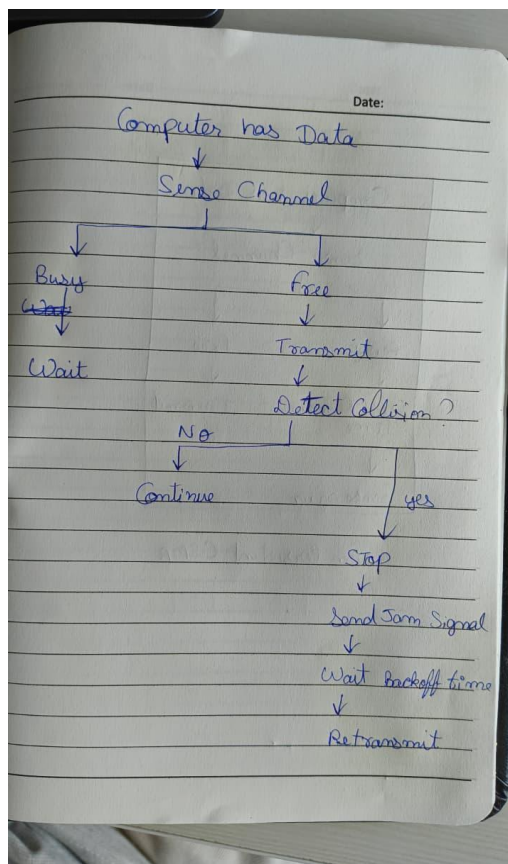
Computer A is mid-transmission when Computer B wants to send. B checks the channel first, finds it busy, and waits for A to finish rather than transmitting blindly — cutting down on collisions compared with ALOHA.

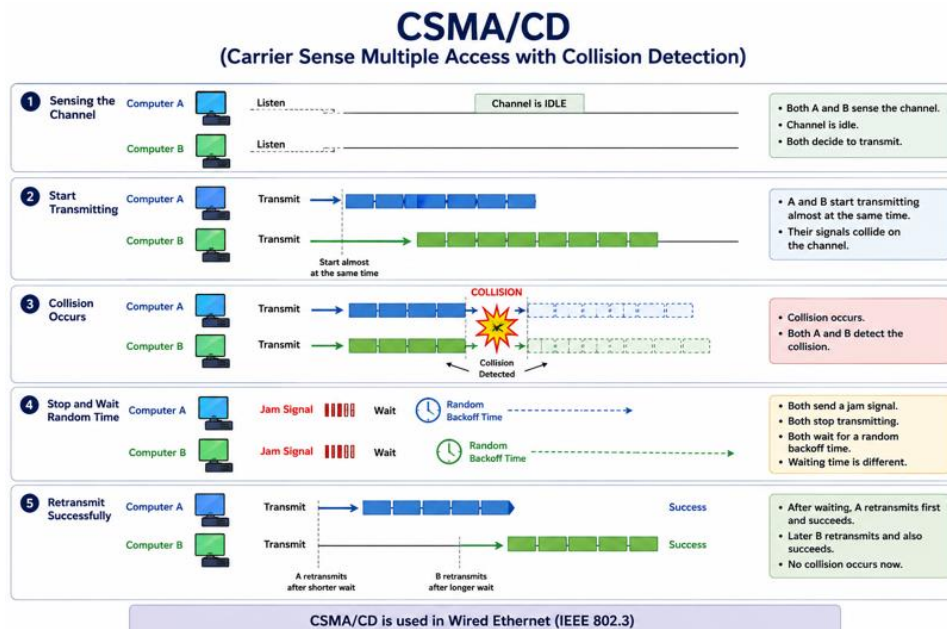
3. CSMA/CD (Carrier Sense Multiple Access with Collision Detection)

CSMA/CD, used on wired Ethernet links, takes CSMA a step further: while it's sending, the station keeps monitoring the channel. The moment it detects a collision, it aborts the transmission right away and puts out a jam signal so every other station on the segment knows a collision just occurred.

Working:

- Sense the channel.
- If it's free, begin transmitting.
- Keep monitoring the line throughout the transmission.
- On collision: stop sending, issue a jam signal, wait a random backoff interval, then retransmit.





Example:

Computer A and Computer B start sending at nearly the same instant. Both detect the collision, halt immediately, and each retransmits after its own randomly chosen wait time.

Where it's used:

- Wired Ethernet — IEEE 802.3

4. CSMA/CA (Carrier Sense Multiple Access with Collision Avoidance)

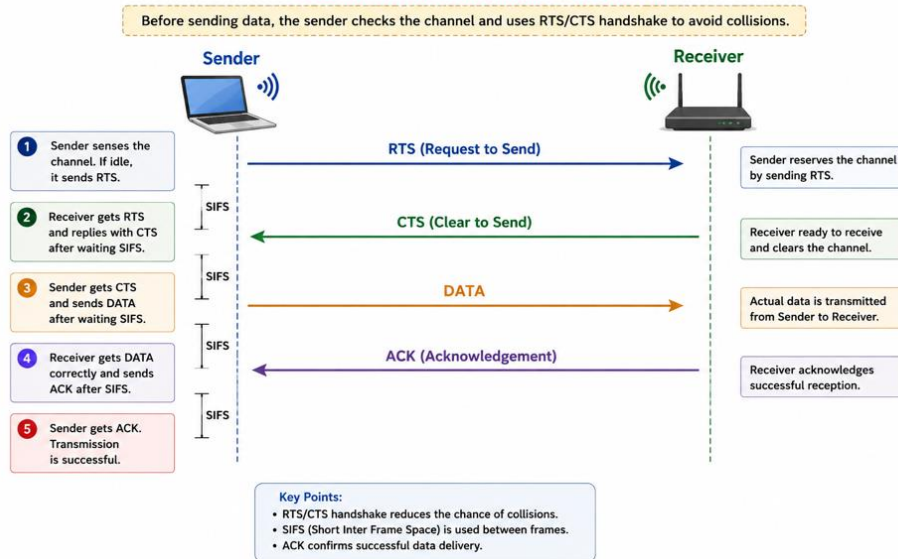
CSMA/CA is the scheme behind Wi-Fi. Wireless stations can't listen and transmit at the same time the way wired ones can, so instead of detecting collisions after the fact, CSMA/CA tries to sidestep them altogether — the sender waits for a clear channel and often exchanges RTS/CTS control frames before any real data goes out.

Working:

- Sense the channel.
- Wait if it's busy.
- Send an RTS (Request to Send) once it's free.
- Wait for the receiver's CTS (Clear to Send) reply.
- Transmit the data.
- Receiver confirms with an ACK.

CSMA/CA (Carrier Sense Multiple Access with Collision Avoidance)

Used in Wireless Networks (IEEE 802.11 Wi-Fi)



Example:

A laptop wanting to talk to a Wi-Fi router first fires off an RTS. The router answers with a CTS, and only then does the laptop send its data — greatly lowering the odds of a collision on the airwaves.

Where it's used:

- Wi-Fi — IEEE 802.11

CSMA/CD vs CSMA/CA

Parameter	CSMA/CD	CSMA/CA
Stands For	Collision Detection	Collision Avoidance
Core Idea	Spot a collision after transmission begins	Steer clear of collisions before sending
Used On	Wired networks	Wireless networks
Governing Standard	IEEE 802.3	IEEE 802.11
Collision Response	Detect, jam, back off, resend	Prevented up front via RTS/CTS
Jam Signal?	Yes	No
RTS/CTS Handshake?	No	Yes
Best Suited For	Wired LANs	Wireless LANs
Typical Example	Ethernet	Wi-Fi